

Kavini Pathagamage

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in kavini-kushani 📧 medium.com/@kavikushi0207 🖨 Kavini Kushani 🏠 1998 - 02 - 07 🇱🇰 Sri Lankan

Objective

A driven fresh graduate who is passionate about exploring new trends and technologies in computer engineering, particularly in cloud applications, telecommunication solutions blended with cybersecurity. I aim to develop innovative and practical solutions to address challenges in data & cloud security and technology sustainability. My goal is to grow as a professional who not only advances in the industry but also contributes to creating a safer and more connected world

Education

Master of Science

Sep 2025 – Present

School of Industrial and Information Engineering, Politecnico di Milano, Italy

- Specialized Area - Computer Science and Engineering
- **Study areas** - Cyber Security, Computer Security, Cloud Computing, Software Engineering, Mobile Application Development, Performance Evaluation with JMT etc.

Bachelor of the Science of Engineering (Honours)

Jan 2019 – Feb 2024

Faculty of Engineering, University of Ruhuna, Sri Lanka

- Specialized Area - Electrical and Information Engineering
- GPA: 3.29/4.0
- **Covered areas** - Mathematics, Programming, Data structures and Algorithms, Machine learning, Object-oriented design principles, Software projects, Telecommunication, Information Security, Computer Network

Diploma in Software Engineering

2018 – 2019

ESOFT Metropolitan Campus, Negombo, Sri Lanka

- Specialized Area - Software Engineering
- **Covered areas** - Basic coding skills, programming languages, CRUD Operations, Database management System, IDE utilization

Professional Experience

Software Engineer Intern

Nov 2023 - Jan 2025

Chimitec S.p.A, Italy

- **Android App:** This is for submitting vehicle defect tickets and tracking updates
- **Web Application:** For ticket management and scheduling maintenance appointments

Research Assistant

Dec 2022 - Oct 2023

Network Softwarization and Security Labs (NetsLab), Ireland

- Developed a **peer-to-peer federated learning platform** for Industrial IoT (IIoT) focusing on privacy preservation
- Researched and implemented privacy-enhancing techniques: Differential privacy, Encryption protocols

Student Instructor

Jan 2023 - Jan 2024

Department of Electrical & Information Engineering, Faculty of Engineering, University of Ruhuna, Sri Lanka

- Contributed to teaching assistance through theoretical and practical examples for undergraduates

Software Engineer Intern

Mar 2022 - June 2022

AdroitLogic Lanka (Pvt) Ltd, Sri Lanka

- Implemented a **customized issue-tracking system** for customer support
- Efficiently redirected tickets to the responsible teams

Publications

Collaborative Federated Learning Platform for Privacy-preserved Decentralized Machine Learning in Industrial IoT at IEEE LATINCOM 2023

[10.1109/LATINCOM59467.2023.10361886](https://doi.org/10.1109/LATINCOM59467.2023.10361886) [🔗](#)

Peer-to-Peer Collaborative Learning Platform for Privacy-Preserved Federated Learning in Industrial Internet - accepted at APAN58 Conference 2024

[10.1007/978-3-031-89813-6_13](https://doi.org/10.1007/978-3-031-89813-6_13) [🔗](#)

Projects

Collaborative Federated Learning Platform for IIoT 2023

- Developed a decentralized federated learning platform for IIoT, including a mobile app for predictions and a secure peer-to-peer data transfer system
- **Technologies:** Java, TCP Protocol, python packages, Distributed system, cryptography, Sockets, React

Online Law consulting firm for Sri Lankan Citizens 2022

- Developed an information portal on Sri Lankan Laws and Chat system for law consultants and clients
- **Technologies:** React, Nodejs, MongoDB, chatbot integration

Distant Patient Care System for COVID-19 2021

- Built user dashboard functionalities for monitoring ICU syringe pumps, infusion pumps, and ventilators to enable remote patient care
- **Technologies:** C#, .NET, SQL, XML

Professional Affiliation

Institute of Engineers Sri Lanka(IESL) 2024 - Present
Associate Member

Institute of Electrical and Electronics Engineers(IEEE) 2019 - 2024
Student Member

Core Competencies and Skills

Technical Skills

- Java, JavaScript, Python
- HTML, Node.js, React, Angular
- Firebase, MySQL, MongoDB
- Windows, MacOS, Linux

Soft Skills

- Leadership
- Team work
- Time Management
- Problem-Solving

Language Skills

- Sinhala (C2)
- English (B2)
- Italian (B1)

Extra Curricular Activities

Positions held

- Secretary of Computer Society(CS) of IEEE - 23'
- University Captain of Powerlifting - 23'
- Chairperson of IEEE HaXtreme - 22'
- Program & Event Organizer of CS of IEEE- 22'
- Faculty Captain of Weightlifting - 22'

Sport Achievements

- University colors for Powerlifting- 23'
- Winner of Weightlifting - 22'
- 1st Runners-Up of Powerlifting - 22'
- 2nd Runner-Up of Shot Put - 22'
- Winner of Power Lifting - 19'

Volunteering

Delegate at IEEE CLAP Region 10 Young Professionals & IEEE SLSWY Congress -22' & 21'